AP CALCULUS AB

Unit 1 • Lesson 1-3 • Notes (filled in)

Lesson 1-3 • Limits to Infinity and Horizontal Asymptotes

Notes (filled in)

Learning goals

- Evaluate limits as $x \to \pm\infty$ for rational, radical and exponential expressions.
- Connect limits at infinity to horizontal asymptotes and infinite limits to vertical asymptotes.
- Compare growth rates of logarithmic, power and exponential functions.

Vocabulary. limit at infinity, horizontal asymptote, infinite limit, vertical asymptote, dominant term.

1. Horizontal and vertical asymptotes

Key idea

If $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$, the line $__y = L__$ is a horizontal asymptote.
If $\lim_{x\to c^-} f(x)$ or $\lim_{x\to c^+} f(x)$ is $\pm\infty$, the line $__x = c__$ is a vertical asymptote.

2. Rational functions

Key idea

Compare the degrees of numerator and denominator: smaller on top $\Rightarrow$ limit $__0__$; equal $\Rightarrow$ ratio of $__$leading coefficients$__$; larger on top $\Rightarrow$ $__\pm\infty__$ (no horizontal asymptote).

Example 1

$$\lim_{x\to\infty} \frac{4x^3 - x}{2x^3 + 7x^2}, \quad \lim_{x\to\infty} \frac{5x + 1}{x^2 - 3}, \quad \lim_{x\to-\infty} \frac{x^3}{x^2 + 1}$$

2. 0. Behaves like x: $-\infty$.

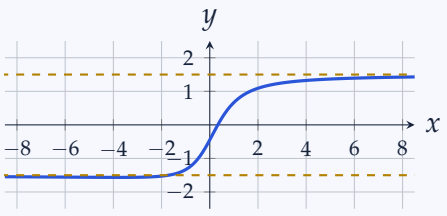
3. Radicals: watch the sign

$\sqrt{x^2} = |x|$, which equals $__-x__$ when $x < 0$.

Example 2

Find both horizontal asymptotes of $f(x) = \dfrac{3x - 1}{\sqrt{4x^2 + 5}}$.

For large $|x|$, $\sqrt{4x^2+5} \approx 2|x|$. As $x \to \infty$: $\frac{3x}{2x} \to \frac{3}{2}$. As $x \to -\infty$: $\frac{3x}{-2x} \to -\frac{3}{2}$. Asymptotes $y = \frac{3}{2}$ and $y = -\frac{3}{2}$.



4. Growth rates

Key idea

As $x \to \infty$: $\ln x$ grows __slower__ than x^p ($p > 0$), which grows __slower__ than a^x ($a > 1$). The fastest-growing term controls the limit.

Example 3

$$\lim_{x\to\infty} \frac{x^3}{e^x}, \quad \lim_{x\to\infty} \frac{3e^x + x^2}{e^x - 1}, \quad \lim_{x\to-\infty} e^x, \quad \lim_{x\to 0^+} \ln x$$

0. Divide by e^x: $\frac{3+x^2e^{-x}}{1-e^{-x}} \to 3$. 0. $-\infty$ (vertical asymptote $x = 0$).

Example 4 • Vertical asymptotes

For $f(x) = \dfrac{x+2}{x^2-4x}$, find the vertical asymptotes and the one-sided limits.

$f(x) = \frac{x+2}{x(x-4)}$. At $x = 0$: numerator near 2; for $x < 0$, $x(x-4) > 0$: $+\infty$; for $x > 0$: $-\infty$. At $x = 4$: numerator near 6; left $-\infty$, right $+\infty$.

Common mistakes

- Forgetting that $\sqrt{x^2} = -x$ for $x < 0$.
- Thinking $\lim_{x\to-\infty} e^x = -\infty$; it is 0.

Check your understanding

1. $\lim_{x\to\infty} \frac{2-5x^2}{x^2+x} =$ __-5__. 2. $\lim_{x\to\infty} \frac{x^{100}}{1.01^x} =$ __0__. 3. $\lim_{x\to-\infty} \frac{2x}{\sqrt{x^2+5}} =$ __-2__.